CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE/IT)

April/May 2012

IT 201 Database Management System

Date: 05.05.2012, Saturday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain the Advantages of DBMS over File Processing System in detail. [05]

(b) Explain the following terms: [02]
1) Serializability 2) Super key

Q - 2 (a) Construct an E-R diagram for Hospital management system. [05]

(b) Explain Aggregation in E-R model with example. [05]

(c) List and explain the levels of Data abstraction. [04]

OR

Q - 2 (a) Construct an E-R diagram for Banking system. [05]

(b) What is DBA? Explain the Functions of DBA. [05]

(c) Explain three levels of ANSI/SPARC architecture of database. [04]

Q - 3 (a) What is normalization? Explain Advantages of normalization. [05]

(b) Define Functional Dependency. Which are the two types of Functional Dependency? What is closure set of FD? [09]

OR

Q - 3 (a) Explain 1NF and 2NF with example. [05]

(b) Explain Armstrong's Axioms. Define and explain canonical cover of FD. [09]

SECTION – II

Q - 4 (a) What is Transaction? Explain ACID Property of transaction. [05]

(b) Explain shadow copy method. [02]

Q - 5 (a) What is concurrency? Discuss concurrency problem with examples. [07]

(b) What is dead-lock? How can we prevent dead-lock? [07]

OR

Q - 5 (a) What are the advantages of concurrency? Explain solutions of concurrency problem with example. [07]

(b) Explain the different types of locking scheme and also explain 2-phase locking protocol [07]

Q - 6 (a) Explain log based recovery in detail. [10]

(b) Explain the aggregate function in SQL with example. [04]

OR

Q - 6 (a) Explain data encryption with example. Explain public key encryption with example [10]

(b) Consider following example and give expression in relational algebra and SQL. [04]

Teacher (T_id, T_name)

Book (B_id, B_name, publisher, price,auther)

Subject (S_id, S_name)

Teaching (T_id, S_id, B_id)

- List the book detail having price more than the average price of subject 'DBMS' Books.
- List the name of all subjects whose books are used by teacher2.

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